Housing First or Treatment First?

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Response to Referee 1

Comments in order, responses in red

Thank you for a report that was detailed enough to rebuild the paper from. You ordered your comments by importance as 2, 1, 4, 3, and the revision has followed that ordering: the quantification was rebuilt first, and it turned up errors in the previous version that we set out below rather than leaving you to find.

1. Model

1(a) Mental health is largely absent

> *"Mental health is central to the economic and policy environment being studied, but it is largely absent from the model. . . . It seems to me that mental health contributing to drug use and addiction as well as homelessness would weaken TF dramatically as a policy."*

Added. Serious mental illness is a persistent binary state that raises the propensity to use, lowers employment, and raises the risk of homelessness. Onset has an age profile estimated from the age of onset distribution, and it is raised by substance use disorder, so the self medication channel runs in both directions. The state is identified by four moments: prevalence in the population, the share of the homeless population with a severe disorder, the unemployment rate conditional on it, and the co occurrence of substance use disorder with it.

Two things are worth flagging.

First, this addition also fixes your comment 2(c)(ii). A model whose only source of persistence is an income process ages people out of homelessness, which is what produced the 73.1 against 48.1 per cent miss at ages 26 to 44. A chronic illness state generates a persistently homeless older population, which is the mechanism the data are pointing at.

Second, we deliberately did *not* add a channel from housing to mental health recovery, and we want to state the reason because its absence looks like an omission. Two independent designs say the channel is not there. The At Home / Chez Soi trial finds housing moves mental illness symptom severity by nothing at any wave, with point estimates of -0.96, -0.51, 0.18 and 0.57, none significant, and moves substance use problem severity by nothing either. Costantini (2025) finds Housing First has *no* effect on mental health care utilisation and attributes his mortality effect to shelter and to safer injection instead. Housing improves mental health *outcomes* in these trials through condition specific quality of life, which the model has no state for, but not through symptom severity or recovery.

On your prediction that mental illness will weaken Treatment First: that is now a testable statement rather than a conjecture, and we report the answer rather than the one we would prefer.

[Built since the first draft of this letter.] Self medication is present, but not as a taste. We considered making the Stone-Geary shifter fall with mental illness and did not, because we have settled on a flow utility that is the same function of consumption, housing and drugs in every state: a preference that moves with a household's own history does the work the mechanism is meant to

do. Mental illness instead lowers the recovery hazard, worsens both employment margins, raises mortality and lowers productivity, all of which raise use without moving the utility function. The onset of mental illness is also no longer a scalar. A flat onset against a flat recovery drives prevalence towards its steady state from below, so the model produced a prevalence profile that *rises* with age where the NSDUH profile falls from 11.0 per cent at ages 26 to 29 to 1.4 per cent at 65 and over. Onset is now backed out of the observed prevalence profile and the recovery hazard, so the data supply the shape and the prevalence target supplies the level.

1(b) Treatment First contains no treatment

> *"One of the goals of TF as a policy is to encourage addicts to seek treatment for their addiction (it is even in the name of the policy!). However, this mechanism is ruled out in the model by the assumption that recovery is random. I found the argument for ruling out any possibility of this mechanism to be fairly weak given its centrality to one of the two policies studied."*

[Conceded.] This was the most damaging sentence in any of the four reports and it was correct. The previous draft's defence, that quitting is largely outside the immediate control of the person with the disorder, is an argument for making treatment *costly and imperfect*, not for making it impossible.

Households with a substance use disorder now choose whether to take up treatment, and take up raises the recovery hazard. The cost is in utility rather than in resources, and the reason is worth stating: a resource cost comes out of the same budget that buys food and rent, so the households with the strongest case for treatment, the poorest, would be exactly the ones who could not afford it. That is an artefact of the modelling choice, not a feature of the world.

The utility cost is identified by treatment take up among people with a substance use disorder, roughly a quarter in NSDUH. This matters because the entire Treatment First result now runs through a parameter that previously had no source at all.

You expect this to strengthen Treatment First relative to Housing First. We report what the model produces.

1(c) The drug price is fixed at one

> *"However, in the model, the drug price appears to be fixed at one and does not adjust in the policy counterfactuals. . . . this seems like a strong assumption that either should be relaxed or better defended."*

Relaxed. The drug price now clears a supply curve, and the Associate Editor and Referee 3 comment M2d ask for the same thing. Referee 3's version is sharper: he points out that one cannot simultaneously normalise the price to one and vary it to compute a price elasticity. He is right, and the normalisation is now explicit about what it does. The price of drugs is the numeraire of the first stage of the calibration, and a scale parameter of the supply curve is renormalised at each market clearing so that the benchmark is invariant to the supply elasticity. The price is then free to move in every counterfactual.

Your point about offsetting is exactly what happens: a policy that reduces the number of users lowers demand and lowers the price, which partially offsets the reduction. The size of the offset depends on the elasticity of drug supply. Because that elasticity is doing real work, we have gone to some length to discipline it rather than assume it. There is no direct estimate for an illicit drug. Cannabis is the closest market to one, because it remains a Schedule I substance federally,

so even a state licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace, Patel and Seegert (2020) estimate a price elasticity of supply of 1.46 on Washington State administrative data, and that is the number the model uses. We do not bracket it with cigarettes or gasoline: those markets share none of those frictions, so their elasticities are not informative about this one.

We also explain why the frequently quoted RAND figure of minus 3.6 is not the right object. A negative supply elasticity reflects enforcement swamping, where a larger market dilutes per unit enforcement risk. Caulkins and Reuter note that swamping "pertains to total drug volume . . . not just or primarily drug specific market volume", so it operates on the aggregate market. Our counterfactuals move one market and leave aggregate volume essentially unchanged, so the force is not active at our margin.

1(d) Show that the mortality channel is the quantitatively important one

> *". . . it seems possible that the labor-market and income-risk consequences of drug use could also generate a low-income concentration of drug use. The paper should demonstrate that the claimed mortality/value-of-life channel is actually the quantitatively important one. A simple way to do this would be to compare the relationship between income and drug use in the baseline model to an otherwise similar version of the model in which mortality is not affected by drug use."*

Done as you specify, and this comment turned out to matter far more than we expected, so we want to be direct about what it uncovered.

There is now a channel shutdown ladder that re solves the model with one channel disabled at a time and reports the income gradient of drug use in each. Your proposed design, mortality not affected by use, is the first rung.

[Conceded.] The uncomfortable part. When we rebuilt the calibration from primary data we found that the previous version was *biased in your favour on both sides at once*:

- The unemployment rate given a substance use disorder was set at 26.9 per cent with no traceable provenance. Measured in the 2024 NSDUH microdata for ages 26 and above, weighted, it is 16.25 per cent. The old figure implied a relative risk of 5.49 against an overall rate of 4.9 per cent; the data give 2.44. The employment channel was running about 2.25 times too strong.
- The homelessness mortality multiplier was a single flat 1.6. Baggett et al. (2013) give race adjusted rate ratios of 8.6 for men and 9.6 for women at ages 25 to 44, 4.5 at 45 to 64 and 1.1 at 65 to 84. The mortality channel was running at roughly a fifth of measured strength, and the age profile was missing entirely.

So the previous calibration overweighted the labour market channel and underweighted the mortality channel. Your rival hypothesis was being handed the advantage by a calibration error, and the paper's own claim was being tested against a version of the model that could not have supported it honestly. Both are fixed, the multiplier is now age varying, and the ladder is run on the corrected calibration.

We also add the diagnostic you say you would find more intuitive than a toy example, and drop the reliance on the static appendix for this purpose (see 1(e)).

1(e) The appendix is disorganised and claims more than it establishes

> *"... the appendix is useful for developing intuition, but I found it somewhat disorganized and narrower than the surrounding discussion suggests. The results are derived in very specific static settings, while the main model is dynamic ... the appendix should be reorganized and made more precise and the main text discussion should describe it more clearly as intuition rather than as a general proof of the model's mechanism."*

Accepted. The static appendix could not establish a claim about continuation values, which is what the paper's mechanism is, so describing it as a proof was overreach.

It is replaced by a stripped down *dynamic* model in the body: two periods, one asset, a binary use choice, a binary housing outcome, and a survival probability that depends on both. Small enough to solve by hand, and it delivers three results the paper actually needs: the sign of the income gradient of use under a mortality penalty against a proportional income penalty; the opposing signs of the value of life and moral hazard effects of housing; and the positive cross partial of welfare in generosity and conditionality, which is the paper's headline stated as a comparative static. The Associate Editor and Referee 3 both ask for the same section.

The old static material is retained in the appendix, tightened, and relabelled as intuition.

2. Calibration and quantification

2(a) Model fit never reported

> *"... a notable omission is the model fit. That is, the author never reports the model's moments even though they would naturally fit in the table. The closest that the author comes to reporting these moments are by saying, 'My benchmark closely matches all calibration targets.' This is incredibly vague."*

[Conceded.] Accepted without qualification, and we agree with the diagnosis in 2(b) that this omission is what made the rest of the quantification impossible to assess.

The calibration table reports target, model value and gap for all 24 targeted moments. Two further moments are held out of the calibration entirely and reported as validation. The phrase "closely matches" does not appear.

2(b) Untargeted misses framed too positively

> *"I find these tables to broadly demonstrating that the model misses substantially across the most important dimensions presented. ... most problematic in my eyes is that the author frames the model as having done a reasonably good job of matching these moments. ... Without being able to verify myself ... I cannot help but question both the fit of the calibration and also the entire quantification exercise."*

Accepted. The framing was not defensible and the new section does not attempt it. Where the model misses, the miss is stated, and the text says what the miss would do to the policy conclusion if it is real.

We would rather report a smaller number of honest validation targets than a large number of moments with favourable framing, so the untargeted section is shorter than the previous one and each entry says explicitly what it is testing.

2(c)(i) Homeless and users are too poor and too detached from work

> *"In the model, all homeless individuals are unemployed, compared with 40 to 52 percent in the data, and average income among the homeless is essentially zero, compared with 19.1 percent of average income in the data. Similarly, the average income of drug users is 41 percent of average income in the model, compared with 74 percent in the data. . . . a model in which these groups are much poorer and less attached to employment than in the data may evaluate HF and TF at the wrong margins."*

The substance of this is right and it is now addressed at the source: the homeless income ratio of 19.1 per cent is a calibration *target* rather than an untargeted moment, and so are the income ratios for drug users and for people with a substance use disorder. A transitory expense shock lets households become homeless without first being driven to the corner of the state space, which is what produced the degenerate income and employment profile you describe.

[**Point of correction.**] On the drug user comparison specifically, there is a population mismatch in the numbers as stated. The figure of roughly 74 per cent is computed over people with a use *disorder*. The model figure of 41 per cent is over people who *use*. Measured consistently in NSDUH for ages 26 and above, the data give 0.951 for users and 0.687 for moderate or severe disorder.

We think targeting both is a sharper test than the one you proposed, and it makes a point the paper wants to make anyway: drug *use* is close to income neutral, and it is *progression* to disorder that concentrates at the bottom. That is the object the mechanism is about. Both are now targets.

2(c)(ii) The model's homeless are too young

> *"In the model, 73.1 percent of homeless individuals are between ages 26 and 44, compared with 48.1 percent in the California data. Conversely, only 9.3 percent of the model's homeless population is age 55 or older, compared with 27.1 percent in the data. . . . If the model shifts homelessness toward younger ages, it may overstate the long-run labor-market and addiction-response channels while understating the role of policies for older homeless individuals."*

Agreed, and the diagnosis of the consequence is right, which is why this mattered more than a distributional cosmetic.

Three changes bear on it. Mental illness supplies a persistent state that does not resolve with age and therefore generates a chronically homeless older population, which an income process alone cannot do. The homelessness mortality multiplier is now age varying, so the model no longer implicitly kills the older homeless at a uniform rate. And the age profile of the homeless population is now reported as a validation moment rather than being folded into a table of successes.

2(d) Match the policy relevant elasticities

> *"Relevant objects include the response of drug use to voucher generosity or conditionality, the response of homelessness to voucher access, the response of labor supply and employment to vouchers, and the response of recovery or addiction persistence to housing status. . . . the RCT literature cited in the introduction may provide identified elasticities that the model can match."*

This is the request that all four reports converge on and it is the core of the new quantification.

The response of recovery to housing status is targeted directly from Milby et al. (2005), whose two arms differ in housing access for homeless adults with a substance use disorder. The contrast between arms identifies the treatment effect; the level in the no housing arm, 18.7 per cent of weeks abstinent, identifies how costly abstention is for a household that already has a disorder. That second parameter was previously asserted rather than measured.

The response of homelessness to voucher access, and of three year mortality to voucher access, are validated against Costantini (2025), untargeted, in the form of a Wald ratio that matches his estimand rather than a hazard calculation that does not. Details in the letter to the Associate Editor under A14.

One tension in this literature is worth stating because it is productive rather than awkward. Milby finds housing raises abstinence; At Home / Chez Soi finds unconditional housing does nothing to substance use. Milby's housing was *contingent on abstinence* and At Home / Chez Soi's was *unconditional*. Contingency is doing the work, not housing. That distinction is exactly what this paper is about, and the two trials together are better evidence for it than either alone.

2(e) The price elasticity moment is not replicable

> *"... I found the description of the calculation of this moment to be extremely vague, to the extent that I would not be able to replicate it. This should be clarified. I would appreciate an appendix describing the model definitions of the moments more thoroughly in general."*

There is now an appendix, `data/DATA_APPENDIX.md` in the replication package, that gives for every moment the exact model object, the data source with its vintage and universe, and the computation that produces the data value from the file. Nothing is described in prose only. It covers the validation moments as well as the targets, and it flags the three places where the universe rather than the arithmetic decides the number, since those are where a reader is most likely to get a different answer from the same source. The package also carries the datasets themselves, so none of it has to be downloaded first.

2(f) Show identification numerically, and show robustness to the elasticity range

> *"I did find the identification explanation in Section 4.2 to be quite helpful. ... I would like this to also be shown numerically. This could be done in many ways: reporting the Jacobian of the moment vector with respect to the parameter vector, reporting the sensitivity statistic from Andrews, Gentzkow, and Shapiro (2017), through a figure showing how perturbing parameters locally around the baseline calibration affects the associated moments. ... take the elasticity of drug use with respect to price which has a range of plausible estimates from -0.36 to -0.9 ... whether or not the results are robust to this choice ... should be demonstrated."*

Both parts. The identification argument is now shown as a perturbation figure: each parameter is moved locally around the calibrated point and the response of every moment is plotted, so a reader can see which moment is informative about which parameter rather than taking our word for it. We report the Jacobian in the appendix.

The price elasticity is re calibrated at the endpoints of your stated range, -0.36 and -0.9, and the policy experiments are re run at each. The results are reported as a robustness table rather than a footnote. The same treatment is given to the other imprecisely measured targets, which is your comment 4(d).

3. Policy analysis

3(a) Loss of consortium is not integrated into the model

> *"If premature deaths impose welfare losses on other individuals, then these losses should in principle affect the value of life and agents' decisions (such as the decision to use drugs). Instead, the paper adds the loss-of-consortium effect only after solving the model. I find this to be an internal inconsistency."*

[Conceded.] Accepted. The adjustment is deleted.

In its place, following Referee 2's comment C2, a disutility from the size of the homeless population enters period utility, where households internalise it as they internalise everything else. The welfare ranking is therefore produced by the model rather than added to it, which is what your objection actually demanded. It is calibrated to a moment rather than chosen, and it gives Referee 3's comment M2c, where is the market failure if use is rational, an explicit answer inside the model.

3(b) The adjustment is ad hoc and a truly conservative value is zero

> *"κ_{cons} was plucked out of thin air, yet makes an enormous impact on the final policy rankings. To play devil's advocate on the point that the chosen value was conservative, a truly conservative value for a parameter plucked out of thin air such as this would be zero, which results in a different policy recommendation."*

[Conceded.] This is unanswerable and we do not attempt an answer. The parameter is gone.

3(c) Tone down the language

> *"Universal TF is not unambiguously preferred under all welfare criteria considered in the paper. . . . An example of language that should be softened is the conclusion's statement that the debate is resolved in favor of universal TF."*

Accepted. The claim that the debate is resolved is removed.

The headline is now a statement about the interaction rather than about a winner: generosity dominates conditionality, and conditionality pays once the benefit being conditioned on is guaranteed rather than rationed. That is a claim about a cross partial, it is what the stylised model in Section 3 proves, and it does not require a welfare ranking between two universal regimes to survive.

Following the Associate Editor's suggestion, welfare and mortality are reported as two dimensions rather than combined through a single parameter, so a reader who weights them differently from us can read off their own ranking.

3(d) Solve proper transition paths

> *"The author should solve for proper transition paths. . . . I find the current strategy, of a several-period PE adjustment, to be a half-measure that is limited in its informativeness and opens up more questions than it answers."*

Done. This became feasible once the housing market general equilibrium was cut at the Associate Editor's request: a transition path now iterates on the labour income tax and the drug price alone.

The path matters for the paper's interpretation of the trial evidence, which is why the half measure was not good enough. Housing responds within a year while substance use disorder adjusts over a decade, so a three year experiment is well powered on housing and badly powered on use, whatever the long run effect. That argument requires an actual path, not an assertion.

4. Sensitivity

4(a) Imperfect detection

> *"TF assumes that drug use is perfectly observed and perfectly enforceable. . . . If drug use were imperfectly observed, this would likely reduce the effectiveness of TF. The paper should analyze the sensitivity of the results to imperfect detection."*

Added, and structured so the previous specification is a corner rather than a special case bolted on. Each voucher holder faces a per period probability of being tested; detection can only occur if a test occurred; and the detection probability conditional on a test is below one. The old model is the point where the testing probability equals one and detection is certain.

4(b) False positives

> *". . . the paper should consider the consequences of false positives. If some non-users lose voucher access because of erroneous tests, then TF may impose substantial costs on individuals who are not violating the sobriety requirement."*

Added as a second parameter in the same kernel. This one is analytically interesting, because it attacks Treatment First at its strongest point: the value of the guarantee. A false positive rate converts a guaranteed benefit into a risky one for households who are complying, which is exactly the property that the paper's cross partial says makes conditionality valuable. We report the rate at which the advantage of universal Treatment First is eliminated.

4(c) Compliance costs, stigma, administrative burden

> *"I found the argument that drug tests would cost only billions of dollars per year to implement to both be most likely a massive underestimation that ignored administrative overhead and assumed far more efficiency than is warranted of government policy, and also to be too large a number to entirely ignore in the first place."*

Both halves of this are right and the footnote is gone. Testing now has a per test cost that enters the government budget constraint, where it competes with everything else the government pays for, and it scales with the testing probability so that a more thorough enforcement regime is more expensive. The sensitivity runs over the cost per test as well as over the testing frequency.

4(d) Sensitivity to hard to identify parameters

"The paper should report sensitivity to parameters that are critical for the policy analysis yet are difficult to identify such as the parameters governing addiction transitions, recovery probabilities, addiction-related mortality, homelessness-related mortality, and the employment consequences of addiction and homelessness."

Done as a sensitivity ladder over exactly this list, and it doubles as the numerical identification evidence you ask for in 2(f).

Two of these parameters are no longer free in the way they were. The homelessness mortality multiplier is now an age profile taken from Baggett et al. rather than a single number, and the employment consequence of substance use disorder is measured in NSDUH rather than assumed. The substance use disorder mortality multiplier is still calibrated internally, to the price elasticity of participation, but it is now reported against the standardised mortality ratio range of 2.5 to 9.8 in Degenhardt et al., so a reader can see whether the level the elasticity chose is consistent with the mortality evidence. That comparison is not a target, and disagreement would be a finding.

Smaller comments and typos

S1 Why a maximum experimentation age?

"I do not understand why there is assumed to be an age at which drug experimentation will not occur. This seems inconsistent with Table 4(a), which shows U-shaped drug use by age."

[Point of correction.] There is a distinction here that the previous draft failed to make clear enough for the objection to land where it should, so we take the blame for the confusion rather than the referee.

The cap is on *initiation*, not on *use*. A household that acquires a substance use disorder at 40 continues to use at 65, so a cap on initiation is consistent with positive and even rising use at older ages. The two objects are different and the previous draft used the word "experimentation" for both.

That said, the underlying discomfort is legitimate: a hard truncation is an unattractive way to represent a smooth empirical regularity, and it was doing work in the calibration. It has been replaced by an age profile in the initiation hazard estimated from the age of onset data, which delivers the same concentration of initiation in early adulthood without a discontinuity, and removes a free parameter.

S2 Tables should show levels and percentage changes

"The results tables would be easier to interpret if they included both percentage changes and baseline/counterfactual levels for the main policy outcomes."

Done. Every results table now reports the benchmark level, the counterfactual level, and the change.

S3 to S10 Typographical corrections

"well-aware" should be "well aware"; equation (9) should end with a comma; remove the comma after the citation in calibration target 19; footnote 7 "our calibration" should be singular; the Abramson

citation lists forthcoming twice; "Kinneruda" and "Kragh-Sørensena" carry spurious trailing letters; Goering et al. has "Guido. Powell ,"; Degenhardt et al. has "Hamamd Ali" for "Hammad".

All corrected. The bibliography has been rebuilt from the publishers' records rather than corrected entry by entry, since the pattern of errors suggested the source file was unreliable. Thank you for reading it that carefully.